

DATAVERSE AFRICA JULY 2026 – HR ANALYTICS CHALLENGE

HR Analytics Dashboard & Workforce Insights Report

Prepared by

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GitHub Repository: <https://github.com/KokoMukruYves/HR-Workforce-Analytics-Excel-Dashboard/edit/main/README.md>

I. Project Overview

This project was developed as part of the **DataVerse Africa July 2026 HR Analytics Challenge**.

The objective was to transform a raw employee dataset into an interactive HR analytics solution capable of supporting evidence-based workforce decisions. The project involved cleaning inconsistent HR records, designing an automated Excel dashboard, and translating workforce data into actionable business insights.

The final solution combines **Power Query**, **Excel formulas**, **PivotTables**, **PivotCharts**, and **interactive slicers** to provide HR managers with a dynamic reporting environment. By simply replacing the raw dataset and refreshing the workbook, all transformations, KPIs, and visualizations update automatically.

II. Dataset Description

The analysis is based on an employee dataset containing **311 employee records** across **39 variables**, covering the period **2006–2019**.

The dataset includes information related to:

- Employee demographics (Employee ID, Gender, Date of Birth)
- Employment history (Hire Date, Termination Date, Department, Manager)
- Compensation (Salary)
- Recruitment channels
- Employee engagement and satisfaction
- Performance evaluations
- Attendance and absence records

Because the raw dataset contained several quality issues, data preparation was completed before analysis to ensure the reliability of all reported metrics.

III. Data Preparation (ELT Process)

The project followed a structured **Extract–Load–Transform (ELT)** workflow.

- **Extract**

The HR dataset was provided as a single Excel worksheet.

- **Load**

The dataset was imported into Excel and processed using **Power Query**, ensuring that all cleaning steps remain fully repeatable and refreshable.

- **Transform**

Several data quality issues were identified and corrected, including:

- Recalculated employee age based on **termination date** or **31 December 2019** for active employees.
- Recalculated employee tenure using the same reference dates.
- Completed missing **ManagerID** values using **XLOOKUP**.
- Standardized text capitalization across categorical variables.
- Corrected inconsistencies in employment status.
- Removed termination reasons assigned to active employees.
- Removed records with invalid employment ages.
- Created additional analytical fields including:
 - Age Group
 - Termination Year
 - Tenure (Years)

Once the data model was validated, an interactive dashboard consisting of KPI cards, PivotTables, PivotCharts, and slicers was developed to support HR decision-making.

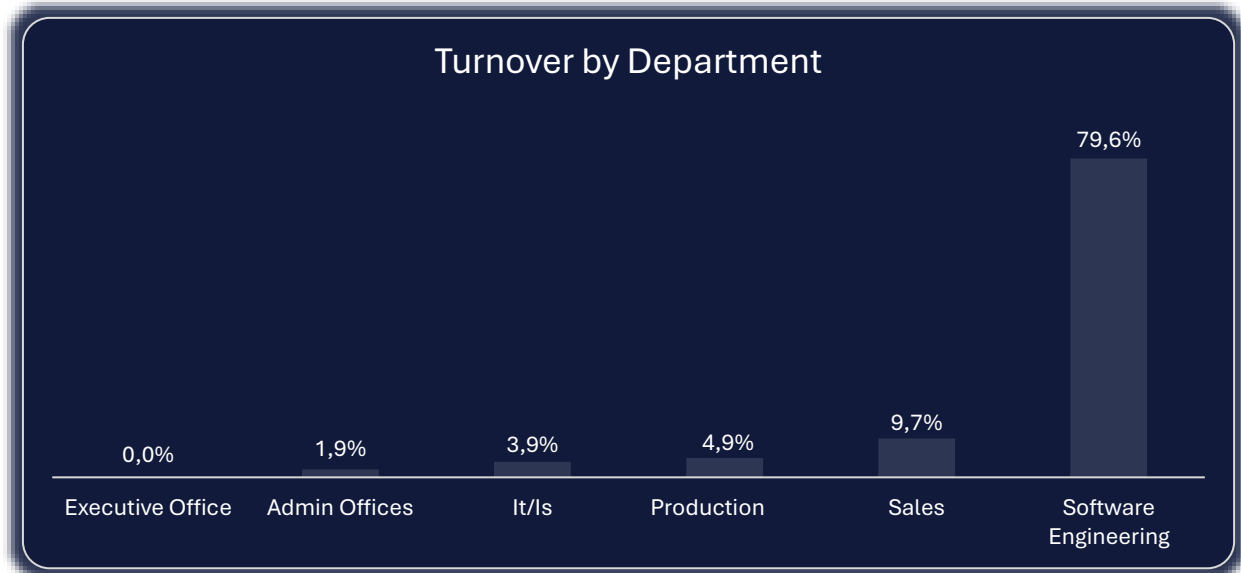
IV. Executive Workforce Summary

KPI	Value
Total Employees	310
Active Employees	207
Terminated Employees	103
Turnover Rate	33.2%
Average Age	39.6 Years
Average Tenure	5.4 Years
Average Engagement	4.1 / 5
Average Satisfaction	3.9 / 5
Gender Ratio (M:F)	1.30
Average Salary	\$69,033.80

Executive Insight

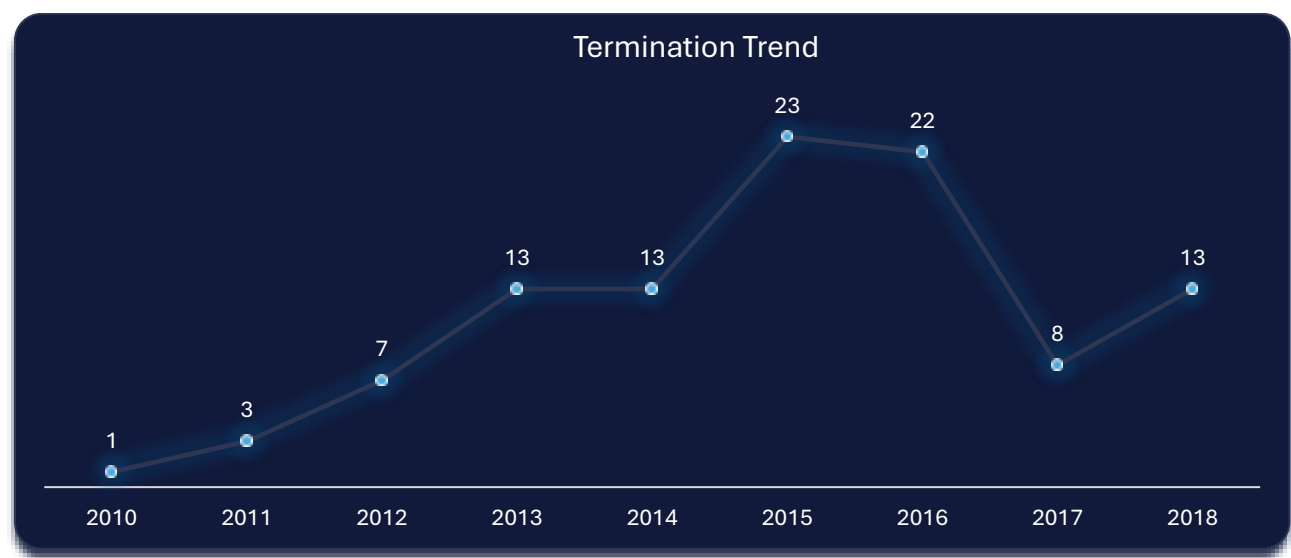
The workforce consists of **310 employees**, with approximately **one-third of employees having left the organization** during the study period. Employee engagement remains relatively high (4.1/5), while overall satisfaction is slightly lower (3.9/5), suggesting opportunities to further improve the employee experience.

Q1. What is the overall turnover rate? Which department has the highest turnover?



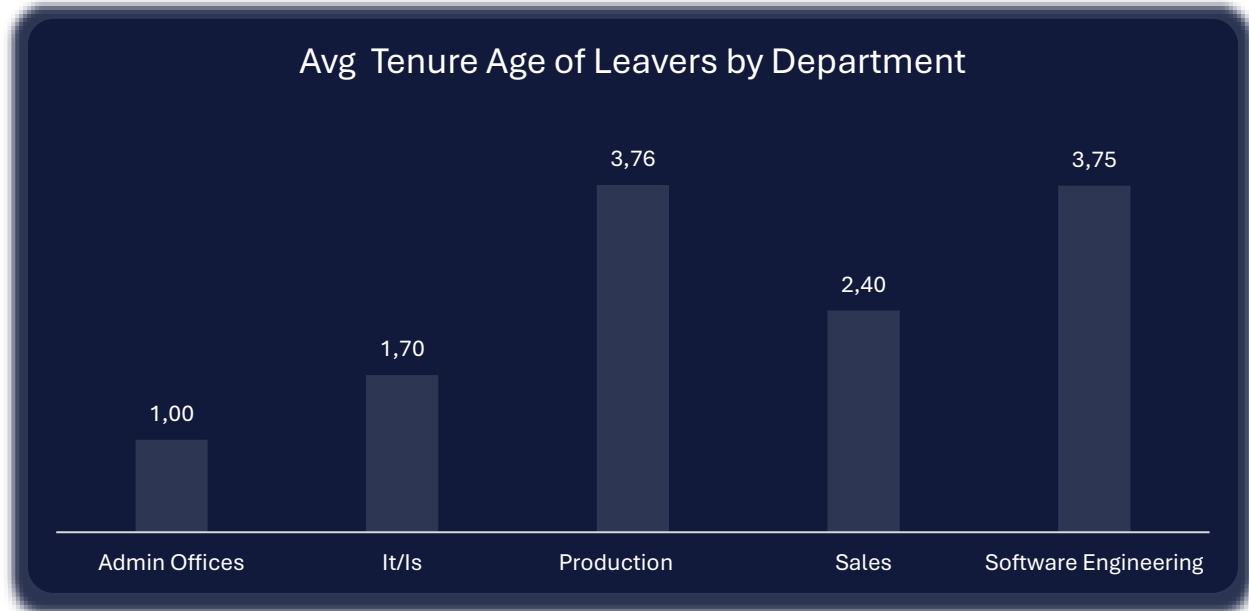
Key Finding	Business Insight	Recommendation
The company recorded an overall employee turnover rate of 33.2%. Software Engineering experienced the highest proportion of employee departures, whereas the Executive Office reported no recorded turnover during the analysis period.	The concentration of turnover within Software Engineering suggests that retention challenges may be more pronounced in technical roles than in administrative functions. While the dashboard identifies where turnover is occurring, it does not explain why employees are leaving.	HR should prioritize a retention review within Software Engineering by analyzing exit interviews, compensation competitiveness, career progression opportunities, workload, and manager effectiveness. Comparing retention practices with departments exhibiting low turnover may reveal strategies that can be adopted across the organization.

Q2. Which year had the highest number of employee terminations?



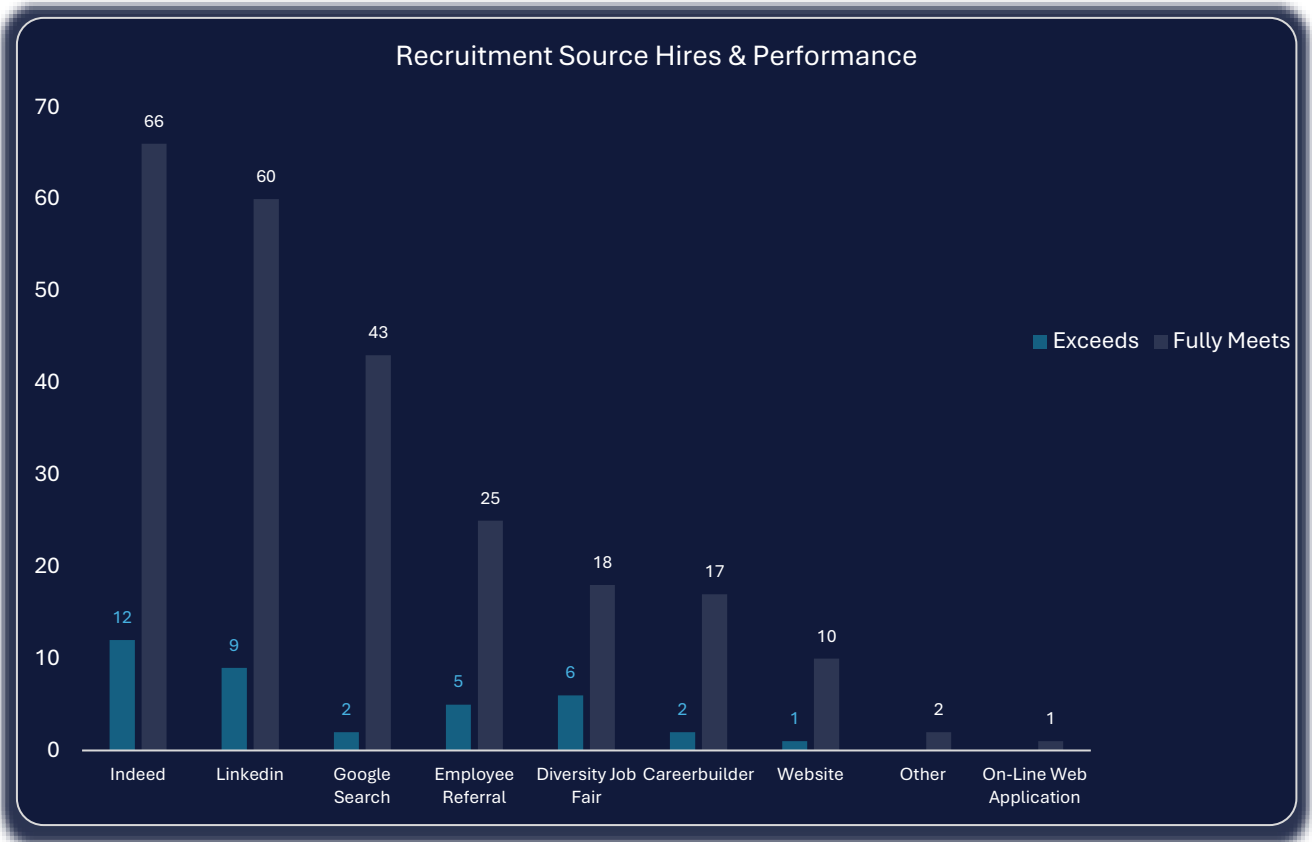
Key Finding	Business Insight	Recommendation
Employee separations peaked in 2016 , with elevated termination levels also observed during 2015 .	A sustained increase in departures across consecutive years may indicate organizational changes, restructuring, shifts in business priorities, or external labour market conditions affecting employee retention.	HR should review organizational events occurring during this period and identify whether common patterns existed among departing employees. Understanding these historical trends can help strengthen future workforce planning and retention strategies.

Q3. Do Production employees leave sooner than IT employees?



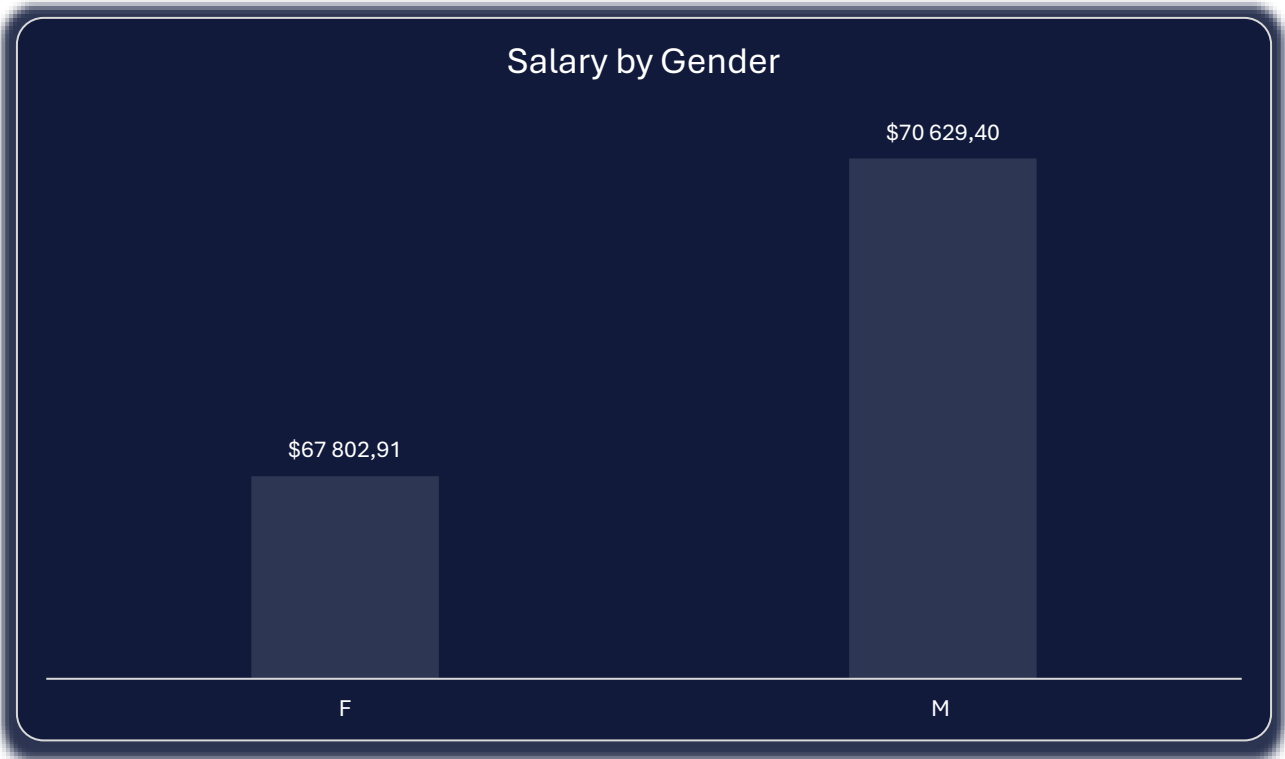
Key Finding	Business Insight	Recommendation
Employees within the IT department demonstrate a shorter average tenure than those working in Production. The overall average employee tenure across the organization is 5.4 years .	Earlier departures within IT may indicate stronger external job opportunities, higher market demand for technical talent, or differences in career progression compared with Production employees.	Retention initiatives for technical talent should focus on career development, continuous learning opportunities, and competitive reward packages designed to reduce voluntary turnover.

Q4. Which recruitment source produces the best employees?



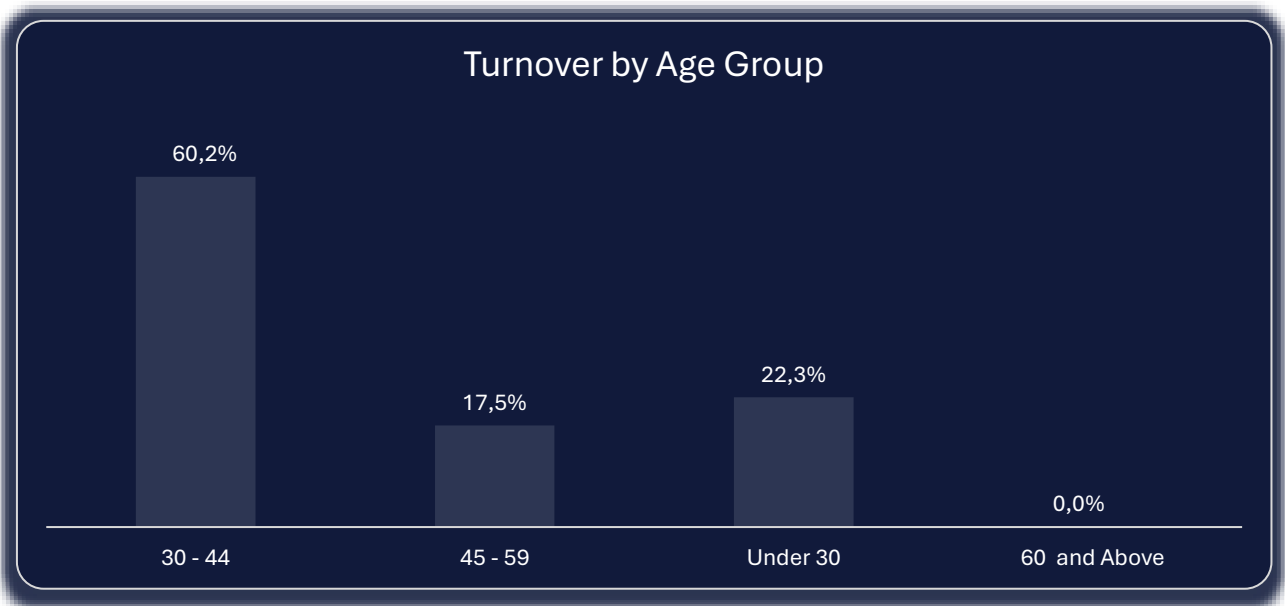
Key Finding	Business Insight	Recommendation
Indeed and LinkedIn generated the largest number of hires while also contributing a substantial proportion of employees rated as Exceeds Expectations .	These recruitment platforms appear effective not only in attracting candidates but also in sourcing high-performing talent.	HR should continue investing in recruitment channels that consistently deliver strong employee performance while periodically evaluating the effectiveness of all recruitment sources using quality-of-hire metrics.

Q5. Is there a gender pay gap? What is the average salary for males vs females?



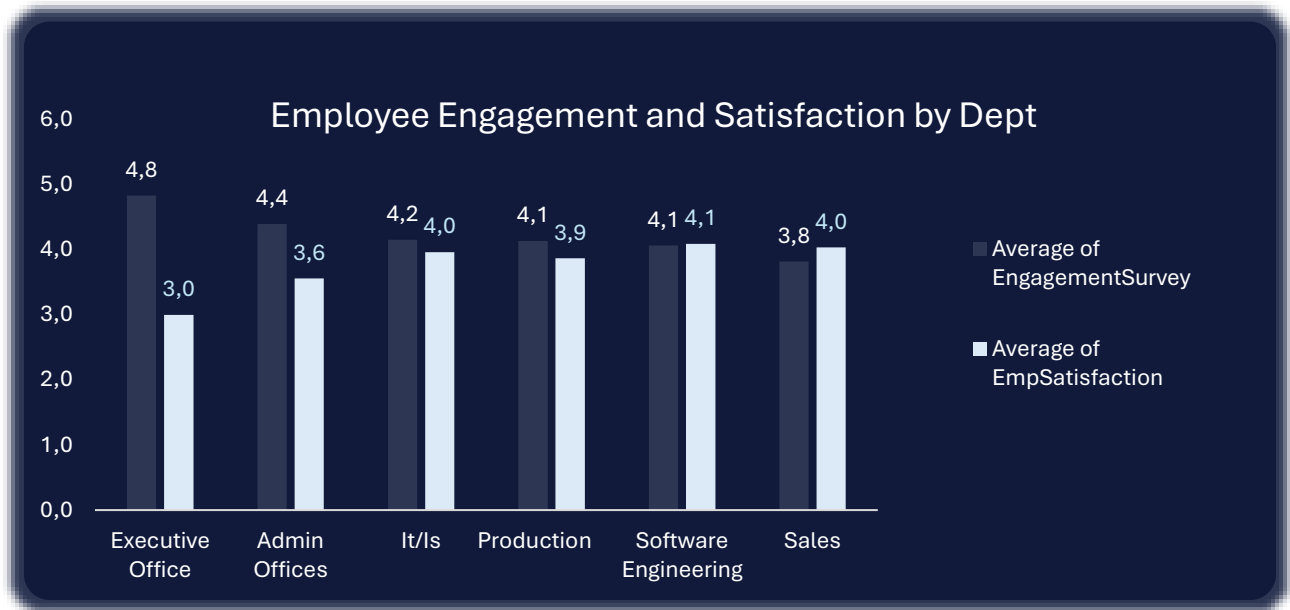
Key Finding	Business Insight	Recommendation
The average salary for male employees is \$71,080.21, compared with \$68,416.48 for female employees. This represents an average salary difference of approximately \$2,664, or around 4%.	While an average salary difference exists, this analysis alone cannot determine whether the difference represents pay inequity because factors such as job level, tenure, department, education, and responsibility were not controlled.	HR should conduct a formal pay equity analysis using comparable job roles and experience levels.

Q6. Which age group experiences the highest turnover?



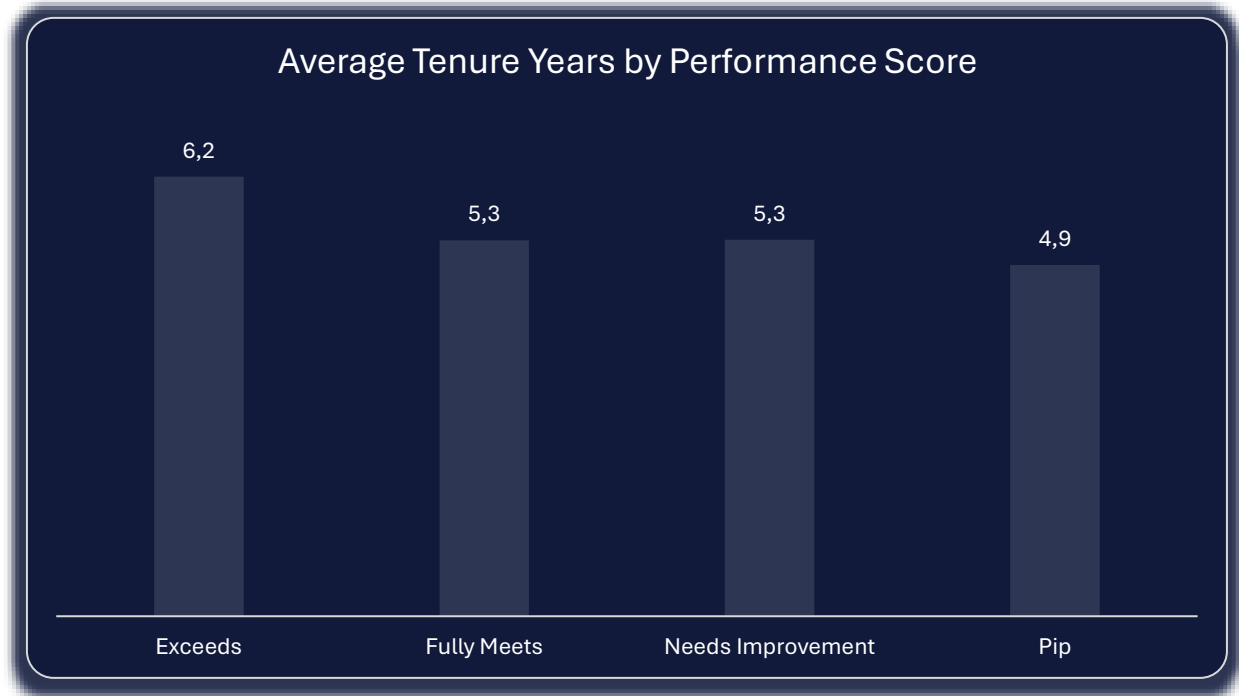
Key Finding	Business Insight	Recommendation
Employees aged 30–44 years account for the largest proportion of employee departures.	This age group typically represents experienced professionals who possess valuable organizational knowledge and transferable skills. Higher turnover among these employees may significantly affect productivity and increase replacement costs.	Retention programs targeting mid-career professionals should emphasize career progression, leadership opportunities, recognition, and flexible working arrangements.

Q7. Which department has the highest average engagement score? Which has the highest average satisfaction?



Key Finding	Business Insight	Recommendation
The Executive Office achieved the highest employee engagement score, while Software Engineering reported the highest employee satisfaction score.	Employee engagement and employee satisfaction do not appear to move identically across departments, suggesting that different organizational factors influence each measure.	Departments with lower engagement or satisfaction should benchmark successful practices from higher-performing departments while considering department-specific operational contexts.

Q8. Do high-performing employees remain longer?



Key Finding	Business Insight	Recommendation
Employees rated Exceeds Expectations remain with the organization for an average of 6.2 years , compared with 5.3 years for employees rated Fully Meets Expectations .	Higher-performing employees demonstrate greater organizational longevity, indicating that employee performance and retention may reinforce one another.	HR should strengthen retention initiatives for high-performing employees through structured career development, succession planning, competitive rewards, and recognition programs to preserve critical organizational knowledge.

V. Tools & Technologies

- Microsoft Excel;
- Power Query;
- PivotTables & PivotCharts for detailed company-level insight;
- Excel Formulas;
- Dynamic Slicers for interactive exploration;

- Figma and Microsoft PowerPoint (for Dashboard Designing);
- AI-generated icons.

VI. Conclusion

This project demonstrates how structured data preparation and interactive dashboard design can convert raw HR data into meaningful business intelligence. Beyond presenting workforce metrics, the dashboard supports strategic decision-making by identifying trends in employee turnover, recruitment effectiveness, compensation, engagement, and retention.

The solution was designed as a fully automated reporting system. Through Power Query, users can replace the source dataset and refresh the workbook to update all transformations, KPIs, PivotTables, charts, and dashboard visuals without manual intervention. This approach makes the dashboard suitable not only as a learning project but also as a scalable template for organizational HR reporting.